



Mutual relationship between surface atmospheric pollutants and CG lightning in Tehran area

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Abstract The mutual relationship between surface atmospheric pollutants and Cloud-to-Ground (CG) lightning is studied in the Tehran area for the first time. The impact of surface pollutant concentration of PM₁₀ on CG lightning initiation, as well as the production of NO₂ and surface ozone derived by lightning occurrence is investigated. To proceed, the reports of thunderstorm occurrence, including CG lightning in the Tehran area have been collected for years of study from the Iran

meteorological organization (IRIMO). The surface pollution data are obtained from the Air Quality Control Center (AQCC) for several stations in the area of interest. The number of lightning (NoL) associated with the selected dates is obtained using the World Wide Lightning Location Network (WWLLN). The correlation coefficients associated with the CG lightning and the concentrations of PM₁₀, NO₂, and surface ozone are calculated. The hourly variations of accumulated NoL and NO₂ and surface ozone are also compared for 24 hours before and after the lightning activity for four days of study during the years of 2009–2013. The results show that there is a positive correlation between PM₁₀ concentration and lightning flash number, obtained from observational data of WWLLN. Moreover, the comparison of NoL and surface pollutant concentration indicates a clear positive contribution from CG lightning in NO₂ and ozone production. In days with a considerable number of lightning occurrences, the comparison of the hourly average of NO₂ and O₃ concentrations with the lightning flash number reveals that NO₂ decreases, and O₃ increases due to the significant increases of lightning strikes.

Highlights

- The first study of mutual relation between atmospheric pollutants and lightning event is presented in Tehran area.
- The number of CG lightning in Tehran area using WWLLN is compared with surface pollutant concentrations taken from AQCC for several stations within five years of study.
- A good correlation is observed between the surface pollutant concentrations and the number of flashes in the area of study.
- The increase of surface ozone and decrease of NO₂ during the lightning activity is observed that shows an agreement with the chemical reactions derived by CG flashes.

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Introduction

Atmospheric pollutants affect the development and formation of local weather systems. The variation of

atmospheric pollutants and their impact on the different stages of weather conditions are not well understood. Many studies have been conducted to explore the physical relationship between lightning occurrence and air pollution. The influence of atmospheric pollutants on cloud systems includes the direct effect on radiation budget as well as the indirect effect on the cloud dynamics and microphysical properties (e.g., Khain et al. 2005; Wang et al. 2011; Li et al. 2017). Hobbs et al. (1974) has shown the weather and climate modification by atmospheric pollutants. Likewise, Bréon et al. (2002) reported the significant effect of aerosols on cloud microphysics on a global scale. There are many studies dedicated to the effect of the urban area on the local weather systems with a focus on the urban heat island circulation and its possible role in air pollution (e.g., Soriano and de Pablo 2002; Steiger et al. 2002; Fallmann et al. 2016). Many studies have been conducted to show the relationship between meteorological conditions and air pollutant concentration (e.g., Khansalari et al. 2020; Yousefian et al. 2020). They empathize with some meteorological factors such as temperature and wind speed as the key influencing factors for surface pollutant concentrations. Mehdipour and Memarianfard (2017) also show that $PM_{2.5}$, PM_{10} , temperature, CO, and NO_2 are the most influencing factors in O_3 prognosticating. Numerous studies have been shown that lightning is a main source of NOx in the middle and upper troposphere and strongly influences NO_2 abundance (e.g., Choi et al. 2005; Ott et al. 2010; Nault et al. 2017). On the other hand, a few studies have been dedicated to the effect of lightning activity on the surface variation of NOx and O_3 (e.g., Middey and Chaudhuri 2013; Kar and Liou 2014; Fei et al. 2016, 2019).

Nitric oxides family NOx (including NO and NO_2) are the primary catalyst for tropospheric ozone production. Intra-cloud and cloud-to-ground flashes are one of the NOx drivers in the atmosphere. While the exact contribution of lightning to the global NOx budget is still the subject of debate in the community, the dominating role of lightning (especially during summer) in controlling middle and upper atmosphere NOx and O_3 variation has been validated (Zhang et al. 2003). NOx emission generated during lightning is proposed to be responsible for 30% of the total NOx emission. Such a contribution to global NOx is expected to impact the ozone peak. While the natural ozone layer in the stratosphere acts as a protective layer of earth from the sun's

harmful rays, bad ozone (surface ozone) which is also known as pollution can be produced in the lower troposphere. More ozone near the surface leads to poor air quality and can cause harm to human health. Subsequent work by Zhang et al. (2014) shows that lightning could enhance surface ozone by about ~10 ppbv during summer and within the 24 hours of the storm. The ozone production during the storm lifetime was estimated originally to be of the order ~2 ppbv. The 3-D simulation using CSCTM (Cloud Scale Chemical Transport Model) of NOx and subsequent tropospheric O_3 production and advection associated with the developing thunderstorm is investigated by Decaria et al. (2005). This work also predicted downstream ozone production of 10 ppbv/day.

Studies show lightning activity is stronger at downwind of the urban area compared to the suburban areas and there is a positive relationship between suspended particle (e.g. PM_{10}) concentration and lightning frequency. This verifies the key role of aerosols in increasing the liquid water content and ice particles in the phase zone of clouds (e.g., Pinto et al. 2004; Kar et al. 2009; Farias et al. 2009).

Considering the substantial increase of air pollution in the second half of the year in the Tehran area, the present study aims at investigating the potential impact of an increase in the concentration of atmospheric pollutants on lightning events as well as their subsequent effect on surface NO_2 and O_3 for the first time. In the present study, the mutual relationship between air pollution and lightning activity in the Tehran area during the years of 2009–2013 is investigated. The pollution and lightning data are presented in second section. Third section provides the correlation coefficients associated with the CG lightning and the concentrations of PM_{10} , NO_2 , and surface ozone. The hourly variations of accumulated NoL and NO_2 and surface ozone are also compared for 24 hours before and after the lightning activity. The hourly correlation coefficients associated with the lightning duration and suppression of surface ozone as well as the increase of NO_2 are discussed. Finally, a summary and conclusion are provided in fourth section.

Data

The location of the current study is the Mehrabad Airport station of the Tehran region with 35.69°N and

51.310E. Thirty days with thunderstorm reports are collected in the Mehrabad Airport station in the Tehran region between 2009 to 2013 based on the current weather code taken from the Iran Meteorological Organization (IRIMO). Then, the number of CG lightning strikes for the selected 30 days of lightning activity is obtained from an area with a 50 km radius from the Mehrabad Airport station from the WWLLN observational network. This network has ground-based receivers in different countries of the world and is capable of detecting lightning events with a current intensity greater than 5 kA. The very low frequency (VLF) radio signal (5–30 kHz) generated by the CG lightning and received at five ground-based receiver stations is used to determine the exact location of lightning. To investigate the relationship of CG lightning with the concentration of surface pollutants in the Tehran area, the daily and hourly averages of PM_{10} , NO_2 , and surface O_3 concentration are obtained from Iran's Air Quality Control Company for three pollution stations including Fath, Shadabad, and Sharif University. Figure 1 shows the geographical location of Mehrabad Airport weather station, along with the locations of the air pollution stations including Fatah, Shadabad, and Sharif. The stations are selected based on the complete available data during the study period as well as spatial compatibility with the WWLLN data. The used methodology for this research is presented in Fig. 2.

Results and discussion

In this section, at first the relationship between the number of lightning flashes (NoL) and the daily average of PM_{10} , NO_2 and O_3 concentrations is studied. Then to obtain the effect of the lightning activity on NO_2 and consequently on the surface O_3 concentration, the variation of hourly concentration data of NO_2 and O_3 and the number of lightning flashes obtained from WWLLN are investigated.

Air pollution effect on lightning initiation

The effect of air pollution effect on lightning initiation is presented in this section. To achieve this goal, the scatterplots of the daily average of PM_{10} concentration and the number of CG lightning flashes obtained from WWLLN are presented in Fig. 3 for the Shadabad and Fatah stations. This Figure shows that PM_{10}

concentration is positively correlated with the number of lightning flashes. In other words, the pollution has led to the intensification of the storm and lightning activity at these stations. The higher aerosol concentration over the surface contributes to more Cloud Condensation Nuclei (CCN) formation, which results in the reduction of the mean cloud droplet size through decreasing the process of coalescence and the droplet collision efficiency. Thus, more supercooled water is likely to exist at greater depths inside the cloud. The abundance of supercooled water may generate large graupel. This could lead to an enhanced lightning activity over the urban regions (Liou and Kar 2010).

On the other hand, to investigate the relationship between lightning activity and NO_2 and O_3 , the scatterplot of the daily average of NO_2 and O_3 concentrations versus the number of CG lightning flashes are shown in Figs. 4 and 5. According to these figures, CG lightning is positively correlated with NO_2 and consequently with ozone, especially at the Sharif station (Figs. 4a and 5a). Note that this amount of NO_2 which is produced by lightning is not the individual source of its tropospheric value and the other sources such as biomass burning, fossil fuel combustion and penetration of stratospheric NO_x have a contribution to it. And so, this lightning-derived NO_2 interacts with the other trace gases in the presence of sunlight and results in the production of O_3 . Therefore, as Fig 5a and b indicate, there is a positive correlation between CG lightning flash number and O_3 concentration.

Lightning effect on surface pollutant concentration

To investigate the effect of lightning activity on surface pollutant concentrations, the hourly variations of the O_3 and NO_2 concentrations along with the NoL have been obtained for each case during the times of lightning activity derived from WWLLN data. The hourly variations of O_3 and NO_2 concentrations on the day of the thunderstorm as well as one day before and after the thunderstorm activity have been presented in Figs. 6a, 7, 8 and 9a for the selected days (30 June 2012, 15 July 2012, 28 September 2012 and 21 April 2013). The surface air pollution data from the Sharif air pollution station has been used in this study. The Convective Available Potential Energy (CAPE) and K-index (KI) of the above-mentioned dates are listed in Table 1. CAPE and K-index values are

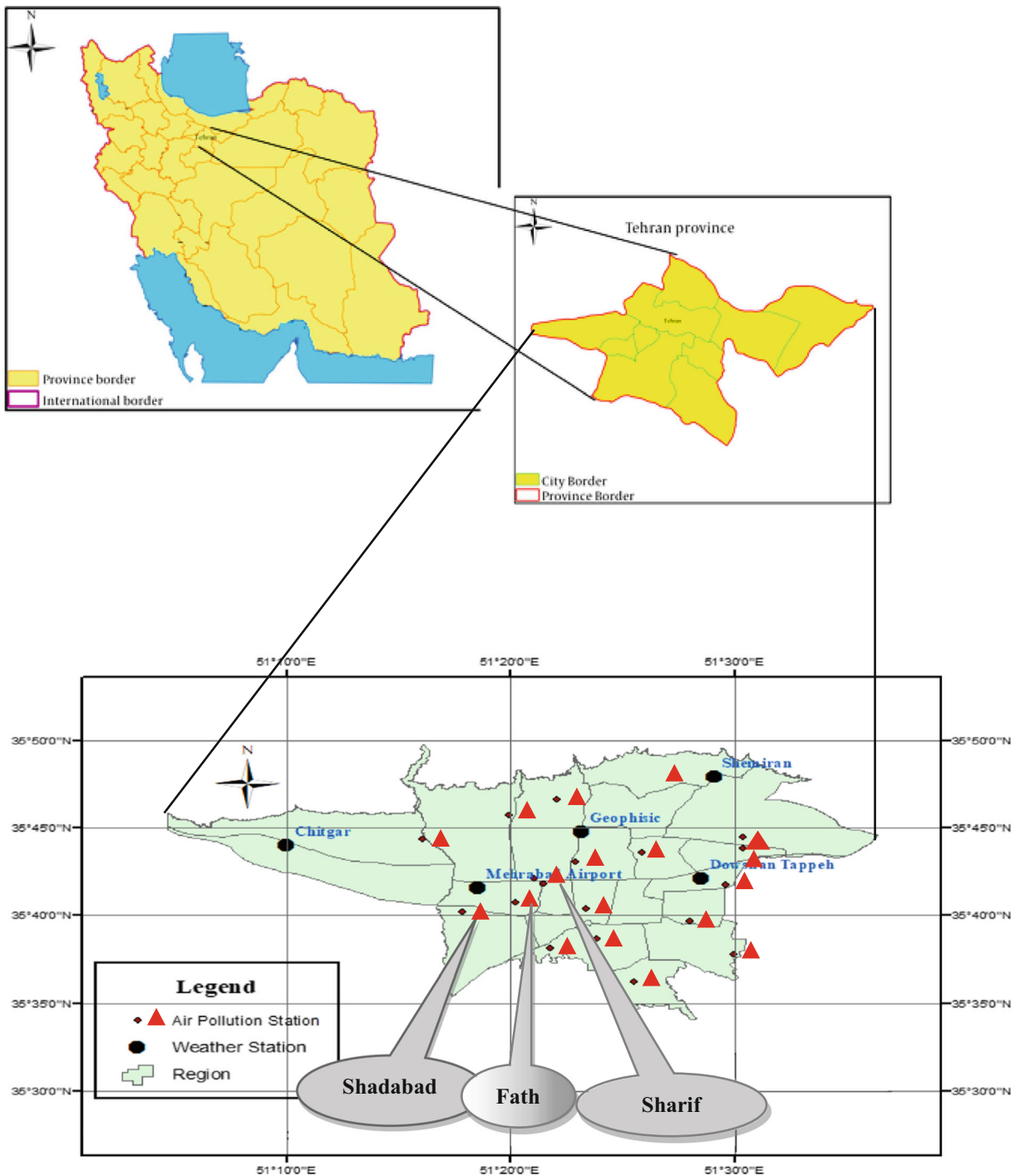


Fig. 1 The map of Iran, the studied city of Tehran and its zoomed including the geographical location of the Mehrabad Airport weather station and the three near air pollution station (Fath, Shadabad and Sharif)

provided by Wyoming University data website (<http://weather.uwyo.edu/upperair/sounding.html>) for the Tehran Mehrabad Airport station.

As shown in Fig. 6b, lightning activity has been reported for 5 hours from 13–18 UTC at the Mehrabad Airport station in the Tehran area on

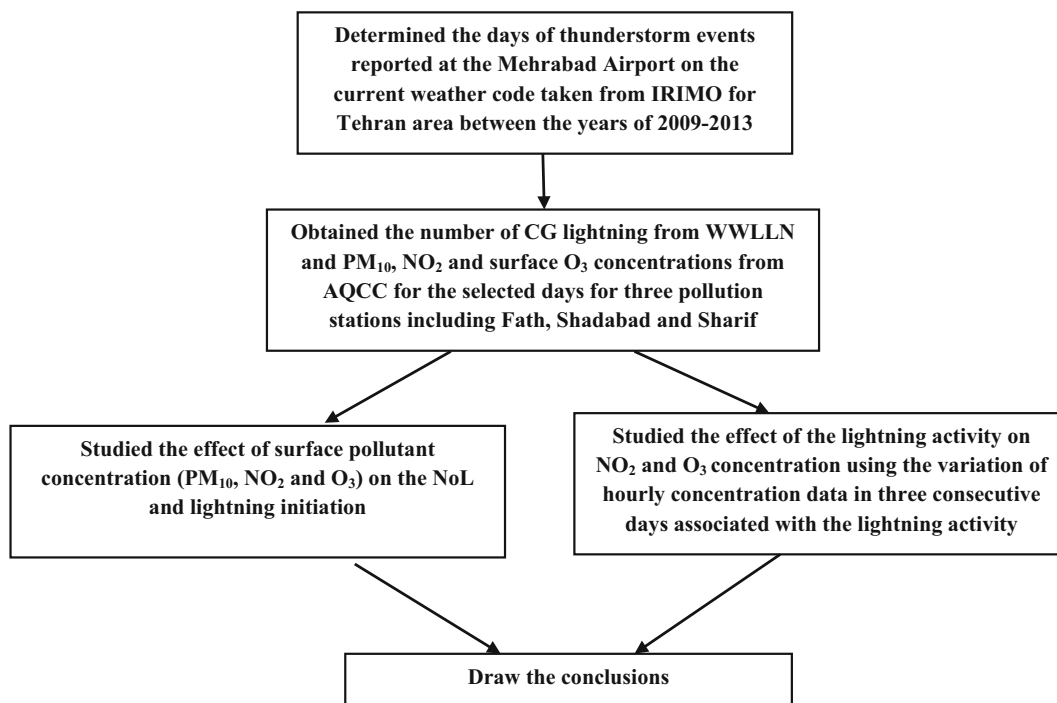


Fig. 2 The used methodology of the research

30 June 2012. On this day, weak lightning activity was observed for 2 hours from 13-15 UTC. The NoL starts to increase afterward and reach 15 flashes at 18:00 UTC. The concentrations of NO₂ and O₃ on the days before and after the thunderstorm day are presented in Fig. 6a. The 72 hours has been considered to be used as a reference to examine pollutants variation from the background levels. A natural increase of O₃ and a decrease of NO₂ during the local daytime (4 UTC-16 UTC)

and due to enhanced chemical reactions in the presence of sunlight. As shown in this Figure, an enhanced O₃ concentration of 40 ppb during the hours of peak lightning activity (16-18 UTC) observed that coincides with the decrease of NO₂ concentration. The detailed variation of this behavior is shown in Fig. 6b. The maximum difference between NO₂ and O₃ reaches ~20 ppb around 16:00 UTC. According to this Figure, the increase in the number of lightning and its positive

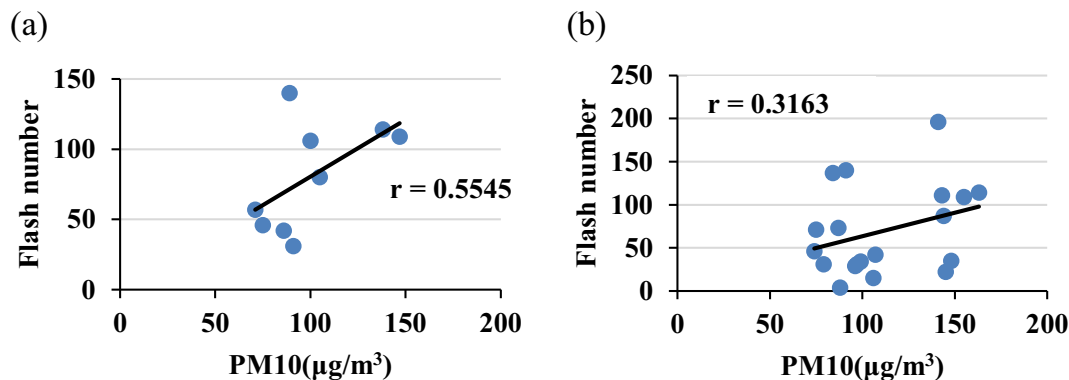


Fig. 3 The scatter plot of daily average of PM₁₀ concentration and the number of CG lightning flashes from WWLLN for the period of 2009-2013 for (a) Sharif station (b) Fatah station

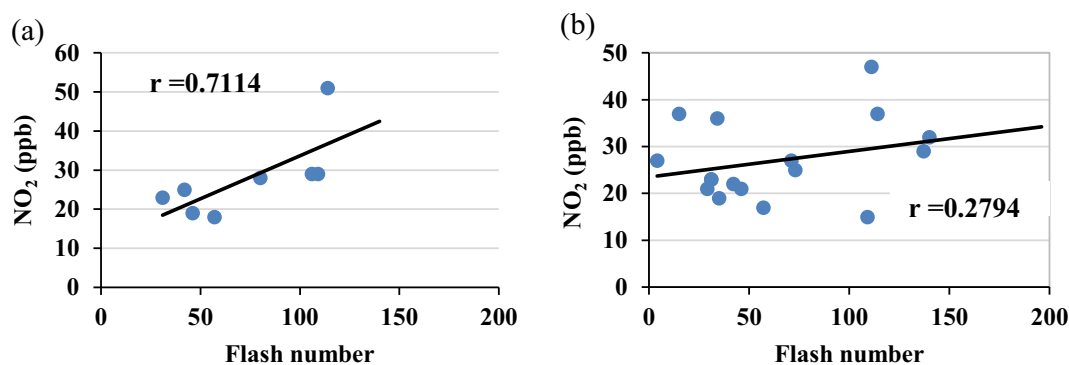
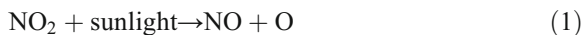


Fig. 4 The same as Fig. 2 but for the daily average of NO₂ concentrations

relationship with O₃ concentration is likely as the result of the below chemical reactions:



which show the consumption of NO₂ and the creation of tropospheric O₃.

There are several other periods on June 30, 2012, that the time evolution of NO₂ and O₃ density that shows the sudden and simultaneous opposite increase and decrease of the two constituents, respectively. This effect could be caused by the convective effects of the storm including downwind formation. The convective transport of either chemical compound could result in the generation of reverse variation of NO₂ and O₃ in contrast to natural behavior in the presence of sunlight and calm weather conditions. Some studies have shown that the effect of NO_x on surface O₃ is complex and nonlinear (e.g., Lin et al. 1988; Pawar et al. 2012). Minschwaner et al. (2008) and Pawar et al. (2012) showed that the decrease of O₃

concentration maybe because of the reduced solar radiation during the active period of thunderstorms.

The above-mentioned effect of lightning activity on the surface NO₂ and O₃ associated with three more days, including a weak and two very strong lightning events is discussed here. The hourly variations of O₃ and NO₂ observed between 14 to 16 of July 2012 including the lightning event that occurred on 15 July 2012 are shown in Fig. 7a. The general behavior of O₃ and NO₂ concentration in the three consecutive days reveals the natural variation during the daytime. As shown in this Figure, O₃ and NO₂ show a significant increase and decrease at the hours of lightning activity between 17–23 UTC, respectively. As mentioned, there is no temporal variation associated with surface pollutants outside the thunderstorm activity, except a smaller amplitude of the formed O₃ on the following day of the event. This could be due to the cloudy conditions and other post-storm activities. This large decrease in NO₂ and the increase in O₃ are obviously due to the two above-

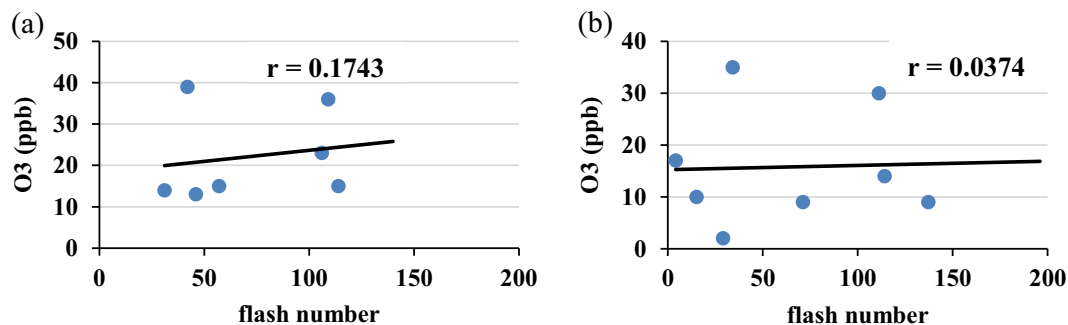


Fig. 5 The same as Fig. 2 but for the daily average of O₃ concentrations

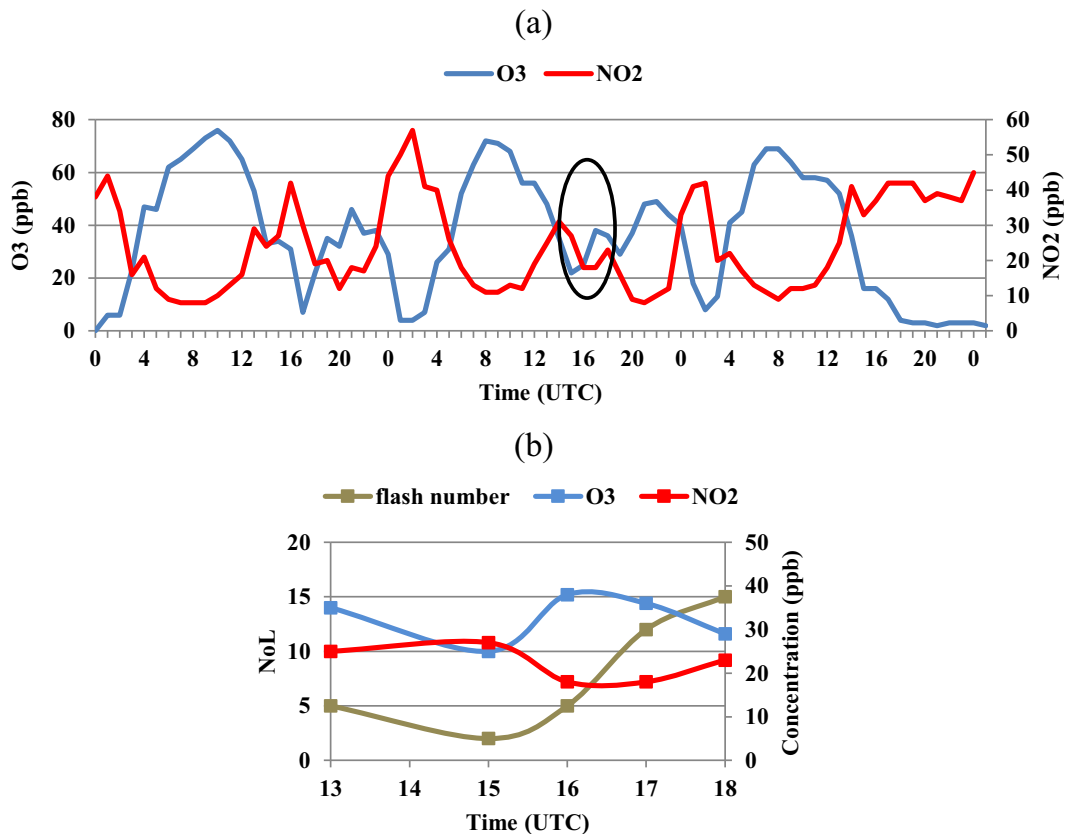


Fig. 6 The hourly variations of **a)** O₃ and NO₂ concentrations in ppb on the day of thunderstorm (30 June 2012) as well as one day before and after the thunderstorm activity day for the Sharif air

pollution station. **(b)** The hourly variations of NoL along with O₃ and NO₂ concentrations in lightning activity hours (black ellipse on plot (a))

mentioned reactions (R1 and R2). In this case, two peaks in the hourly variations of O₃ concentration (Fig. 7b) during the lightning activity hours; between 17–19 UTC and 19–23 UTC, respectively, are observed. The NoL is increased during these hours. The maximum difference between the two constituents reaches ~40 ppb around 21:00 UTC that lies within the maximum lightning activity. It is noteworthy that the second and more intense peak in the hourly variations of O₃ concentration occurred during the dissipation stages of the thunderstorm which indicates the transport of NO_x from the upper or middle troposphere with the downdraft of the thunderstorm during its dissipation stage and the creation of tropospheric O₃ according to R1 and R2 chemical reactions.

The hourly variations of O₃ and NO₂ on the third case study of 28 September 2012 are shown in Fig. 8a including the variation in three

consecutive days including the storm day. The main characteristic that can be observed in this figure is the time evolution of surface pollutants with the daytime and increase of sunlight. The chemical reactions including a normal increase of O₃ during sunlight and subsequent suppression of NO₂ due to the photochemical process are pronounced. The NoL of this case is considerable in comparison with the first two cases and reach about 50 strikes within the region of interest around 21:00 UTC. As shown in Fig. 8b, there is a significant peak in the hourly variations of O₃ concentration during the dissipation stages of the thunderstorm at 21:30 and during the lightning activity.

The fourth case study on 21 April 2013 shows weak lightning with an average NoL of less than 6. While the difference between maximum O₃ enhancement and NO₂ decrease is about zero, a

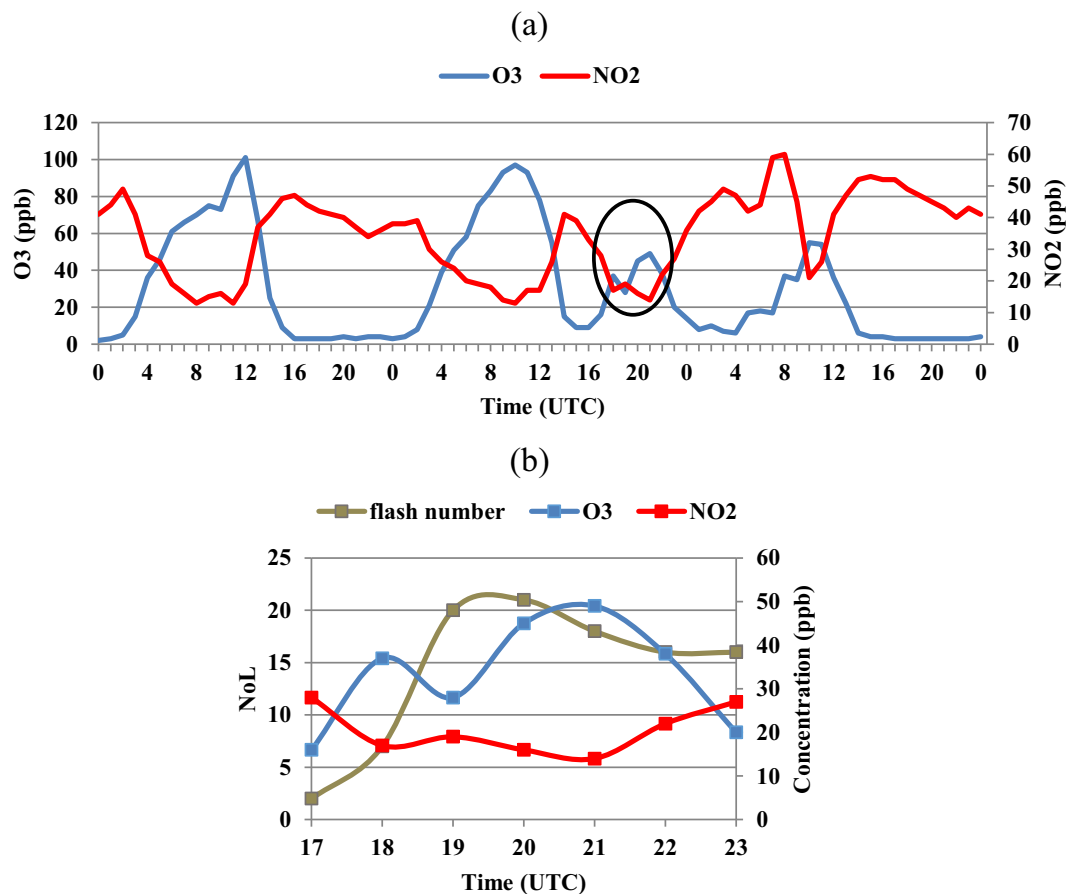


Fig. 7. The hourly variations of a) O₃ and NO₂ concentrations in ppb in three consecutive days associated with the lightning activity on 15 July 2012 for the Sharif air pollution station. (b) The hourly

variations of NoL along with O₃ and NO₂ concentrations in lightning activity hours (black ellipse on plot (a))

clear respected variation can be seen between 14–15 UTC. It is observed the decrease in NO₂ and the increase in O₃ occurs during the damping of the dissipation stages of the thunderstorm (Fig. 9a and b). The simultaneous increase/decrease of O₃/NO₂ is observed throughout this day. As discussed before, this is most likely due to the complicated convective pattern of the storm and the formation of downwind structures.

To elaborate on the simultaneous variation of surface ozone produced as well as NO₂ distinction during the CG lightning, the correlation coefficient with NoL during activity hours is calculated for the 4 events presented in this paper. The case study of 30 June 2013 during CG lightning hours of 3–18 UTC, which corresponds to the local afternoon time, shows a correlation coefficient of

0.63 and -0.73 with O₃ and NO₂, respectively. The NoL associated with this case was 39. The cases of 15 July and 28 September 2012 occurred around 17–23 UTC corresponding to night time with strong lightning activity (NoL = 106, 100) reveals a correlation coefficient of (0.5, -0.53) and (0.82, -0.73) with O₃ and NO₂, respectively. Although the two lightning activities occurred during the same daytime, the calculated correlation coefficients are different. Based on the results presented in Table 2, it can be argued that lightning activity during daytime hours shows a weaker correlation with the production of surface ozone and a decrease of NO₂. A more solid conclusion on the correlation level between NoL and the surface pollutant concentration of O₃ and NO₂ requires more cases to be included. Such a

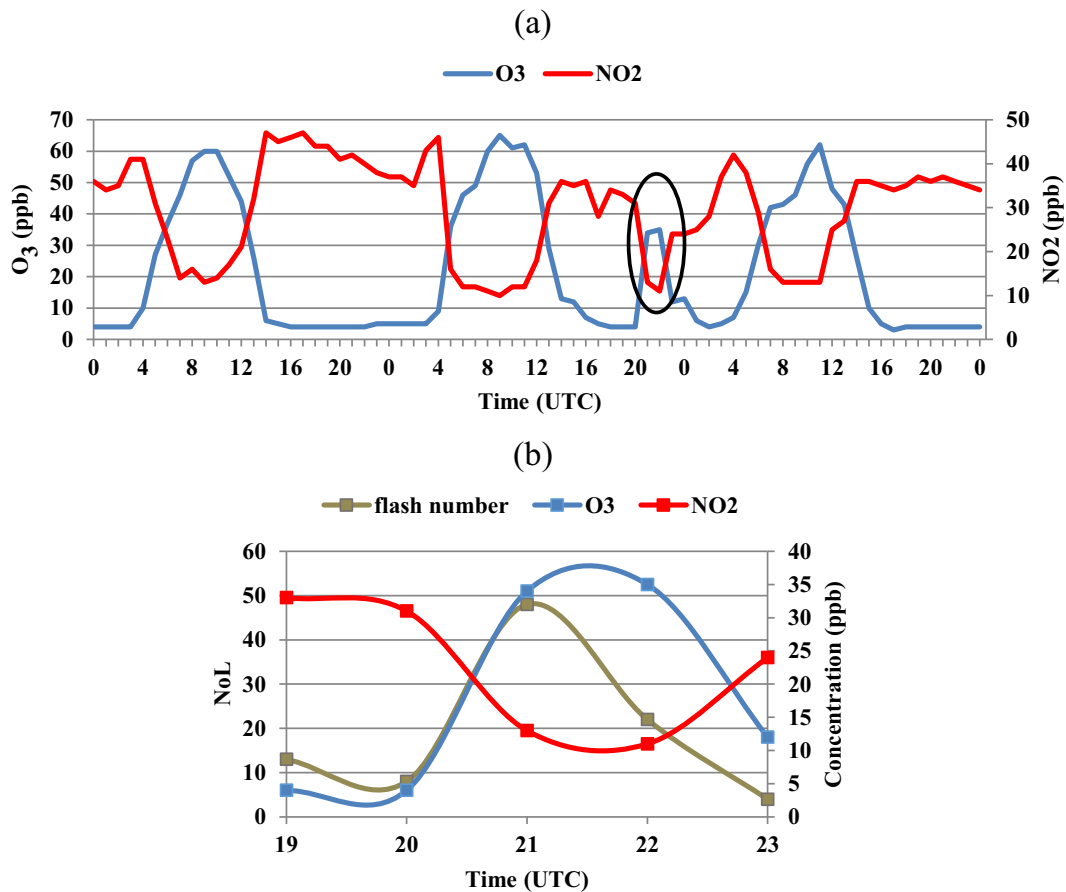


Fig. 8. The hourly variations of **a)** O₃ and NO₂ concentrations in ppb in three consecutive days associated with the lightning activity on 28 September 2012 for the Sharif air pollution station. **(b)** The

hourly variations of NoL along with O₃ and NO₂ concentrations in lightning activity hours (black ellipse on plot **(a)**)

relationship could shed light on possible chemical reactions associated with CG lightning.

Summary and Conclusion

The mutual relationship of air pollution on the lightning generation and the follow-up process of NO₂ and O₃ modification near the surface as a consequence of the CG lightning strike is investigated for the first time in the Tehran area. The relationship between the lightning generation and the increase of pollutant constituents such as PM₁₀, NO₂ is investigated using the available data between 2009 and 2013. The positive correlation between the increase of pollution and intensification of the storm accompanied by lightning is

observed in the Tehran area. This has been attributed to the formation of aerosol that acts as Cloud Condensation Nuclei (CCN) and can cause the suppression of the cloud droplet size. Enhanced supercooled water inside the cloud lead to large graupel generation and an increase in lightning initiation.

The hourly variation of surface NO₂ and O₃ during lightning activity is examined for the day of thunderstorms as well as the day before and after the lightning event. The correlation coefficients of the NoL with NO₂ and O₃ show the decrease of NO₂ concentration and increase of O₃ density as a result of chemical processes initiated with CG flashes. The night-time CG lightning activity revealed a stronger correlation. A more comprehensive study is required to determine the

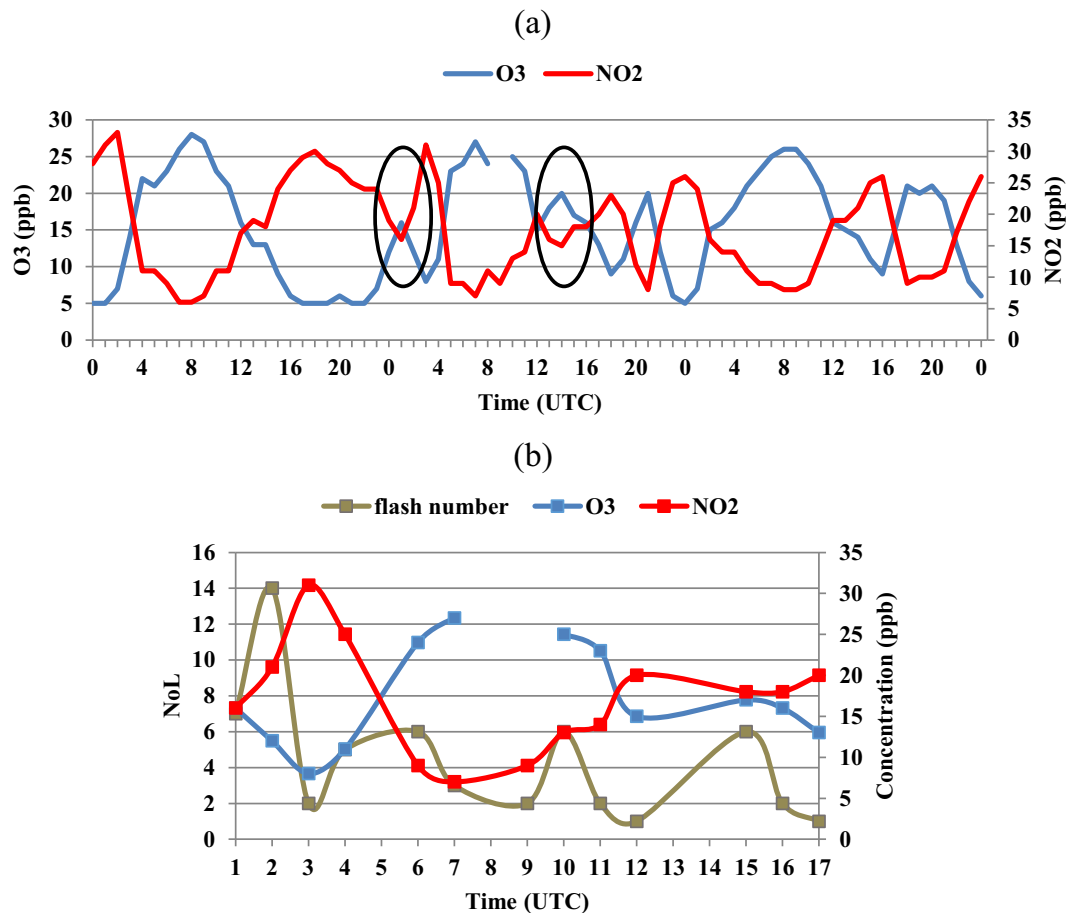


Fig. 9. The hourly variations of a) O₃ and NO₂ concentrations in ppb in three consecutive days associated with the lightning activity on 21 April 2013 for the Sharif air pollution station. (b) The hourly

variations of NoL along with O₃ and NO₂ concentrations in lightning activity hours (black ellipse on plot (a))

exact relationship between chemical reactions, NoL, day-time, and season.

The regional extent of the convective systems and trajectories of the cloud outflow will be investigated in future work in order to determine the significant contribution of O₃ per day produced as the outflow is transported downstream for several days. To achieve this goal, thunderstorms with

high CAPE will be selected along with several stations to monitor the time evolution of upper-tropospheric production within a few days of storm initiation. Such ozone variation in the upper troposphere and its relationship with the lightning NO₂ production should be easily resolved using Dobson observations (~10ppbv is equivalent to ~1 Dobson Unit).

Table 1 The CAPE and KI values for each case study

Case study	CAPE (J kg ⁻¹) at 12:00 UTC	K-index at 12:00 UTC
30 June 2012	1211	41.30
15 July 2012	485	40.46
28 September 2012	0	17.10
21 April 2013	44.16	27.10

Table 2 The hours of lightning activity and the correlation coefficients between NoL, O₃ and NO₂ along with the total number of flashes at this time for each case of study

Case study	Hours of lightning activity (UTC)	The correlation coefficient between NoL and O ₃	The correlation coefficient between NoL and NO ₂	Total number of flashes
30 June 2012	13–18	0.63	-0.73	39
15 July 2012	17–23	0.53	-0.5	106
28 September 2012	17–23	0.82	-0.73	100
21 April 2013	3–7	0.31	-0.40	16
	10–17	0.55	-0.54	18

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Compliance with ethical standards

Conflict of interest The authors certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

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